

Discovering First-Order Logic in Arbitrary Embedding Spaces: Trajectory Displacement Analysis of Latent Relational Structure

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Abstract

Existing neurosymbolic approaches engineer embedding spaces to support logical operations -- training relation-specific transformations (TransE), designing geometric constraints (box embeddings), or encoding logical axioms into loss functions (Logic Tensor Networks). We take the opposite approach: given a general-purpose embedding model with no logical training signal, we systematically discover which first-order logical operations are latently encoded as vector arithmetic. Our method embeds entities from a knowledge base (Wikidata), computes displacement vectors for each predicate's triples, and tests whether these displacements are geometrically consistent -- i.e., whether applying the same vector operation to different entity pairs produces parallel results. On 41,725 embeddings from mx-bai-embed-large (1024-dim), we discover 86 predicates that function as consistent vector operations (alignment > 0.5), with 32 achieving strong consistency (> 0.7). The strongest operations achieve perfect prediction (MRR = 1.0) via leave-one-out vector arithmetic, and two-hop composition succeeds at 28.3% Hits@10. Crucially, we find a 0.78 correlation between geometric consistency and prediction accuracy, meaning the discovery metric itself predicts which operations will work. The method requires no training, no gradient optimization, and no assumptions about the embedding geometry -- it is a pure diagnostic applied to an existing space. We characterize failure modes: symmetric relations (siblings: 0.026), temporally ordered sequences (follows: 0.050), and semantically diverse predicates (instance-of: 0.244) resist vector encoding, revealing that embedding spaces encode functional relationships as geometry but not relational ones. All code and data are publicly available, and the analysis reproduces end-to-end in approximately 30 minutes on commodity hardware.

1. Introduction

A persistent question in representation learning is whether embedding spaces encode logical structure, and if so, how faithfully. The word2vec analogy king - man + woman $\approx$ queen (Mikolov et al., 2013) demonstrated that at least some relational structure exists as vector arithmetic in distributional embeddings. TransE (Bordes et al., 2013) formalized this insight for knowledge graphs, training embeddings such that $h + r \approx t$ for each triple (head, relation, tail). Subsequent work introduced rotations (RotatE; Sun et al., 2019), complex-valued embeddings (ComplEx; Trouillon et al., 2016), and geometric constraints for hierarchical relations (box embeddings; Vilnis et al., 2018).

All of these approaches share a common assumption: the embedding space must be constructed with logical operations in mind. The relation vector r in TransE is a learned parameter, optimized specifically to make $h + r \approx t$ hold. The rotation in RotatE is a learned transformation. The box boundaries are learned constraints. If you take a general-purpose text embedding model -- one trained for semantic similarity, not for logical reasoning -- none of these guarantees hold.

Our contribution is empirical rather than architectural. We ask: given an embedding model that was never trained for logic, which logical operations does it already encode? We introduce trajectory displacement analysis, a model-agnostic method for discovering latent logical operations in any embedding space. Given a knowledge base of ground-truth triples, we discover the subset of predicates whose triples manifest as consistent vector displacements -- treating the embedding space as existing infrastructure to be analyzed, not a system to be constructed. No training is required. No parameters are learned. The method is a pure diagnostic applicable to any text embedding model.

1.1 Key Findings

1. 86 discovered operations. Of 159 predicates tested (≥ 10 triples each), 86 produce consistent displacement vectors (mean alignment > 0.5 with the predicate's mean displacement), and 32 achieve strong consistency (> 0.7).
2. Prediction via vector arithmetic. The strongest operations predict unknown triples with perfect accuracy (MRR = 1.0 for demographics-of-topic, culture, economy-of-topic) using leave-one-out evaluation. Across all 86 operations, mean Hits@10 = 0.550.
3. Composition works. Chaining two displacement vectors ($S + d_1 + d_2$) predicts two-hop targets at 28.3% Hits@10 over 5,000 tested compositions, with successful chains like Fujiwara no Tadahira $\rightarrow$ [citizenship] $\rightarrow$ Japan $\rightarrow$ [flag] $\rightarrow$ flag of Japan (Rank 1).
4. Consistency predicts accuracy. The correlation between a predicate's displacement consistency (alignment) and its

prediction quality (MRR) is $r = 0.78$ ($p < 0.001$). This means the discovery metric itself tells you which operations will be reliable -- without needing ground-truth labels for evaluation.

5. Functional vs. relational split. Operations succeed when the predicate is functional (many-to-one or one-to-one: flag, central bank, head of state) and fail when the predicate is relational (many-to-many or symmetric: sibling, spouse, shares-border-with). This is not a limitation of the method -- it reveals what embedding spaces actually encode.

2. Related Work

2.1 Knowledge Graph Embedding

TransE (Bordes et al., 2013) models relations as translations: $h + r \sim t$. Our work tests whether this property emerges in spaces not trained for it. RotatE (Sun et al., 2019) and ComplEx (Trouillon et al., 2016) use richer transformations but still require training. Our method uses no learned parameters.

2.2 Word Embedding Analogies

Mikolov et al. (2013) showed that $\text{king} - \text{man} + \text{woman} \sim \text{queen}$ holds in word2vec. Subsequent work (Linzen, 2016; Rogers et al., 2017) showed these analogies are less robust than initially claimed, often reflecting dataset biases rather than true relational understanding. Our approach is more systematic: instead of testing cherry-picked analogies, we exhaustively test all instances of each predicate and report aggregate statistics.

2.3 Neurosymbolic Integration

Logic Tensor Networks (Serafini & Garcez, 2016), Neural Theorem Provers (Rocktaschel & Riedel, 2017), and DeepProbLog (Manhaeve et al., 2018) integrate logical reasoning into neural architectures. These are constructive approaches: they build systems that can reason logically. Our approach is analytical: we examine an existing system to discover what logical reasoning it already supports.

2.4 Probing Approaches

Probing classifiers (Conneau et al., 2018; Hewitt & Manning, 2019) test what linguistic properties are encoded in learned representations. Our method is analogous but operates at the relational/logical level rather than the syntactic level, and uses vector arithmetic rather than learned classifiers -- making our results directly interpretable as geometric properties of the space.

2.5 Vector Symbolic Architectures

Vector Symbolic Architectures (VSAs) perform algebraic operations -- binding, bundling, permutation -- on high-dimensional vectors to represent symbolic structures [CITATION NEEDED -- Kanerva, 2009; Plate, 2003]. VSAs are primarily applied in cognitive science and neuromorphic hardware, constructing hypervector representations from scratch. Recent work has begun probing LLM internal representations using VSA-inspired methods [CITATION NEEDED -- "Hyperdimensional Probe"]. Our approach differs in a key respect: VSAs construct symbolic algebras over vectors, while we discover what algebraic structure already exists in embedding spaces not built for it. We do not impose a binding operation -- we test whether the space's native geometry already functions as one.

Separately, work on embedding space topology has identified stratified sub-manifolds of different dimensions within learned representations [CITATION NEEDED -- stratified manifold work]. This independently validates the intuition behind our three-regime classification (Section 5.3): embedding spaces are not uniformly structured but contain regions of varying representational density and reliability.

3. Method

3.1 Problem Formulation

Given:

* An embedding function $f: \text{Text} \rightarrow \mathbb{R}^d$ (any text embedding model)

* A knowledge base $\mathcal{K} = \{(s, p, o)\}$ of subject-predicate-object triples

Find: The subset of predicates $P^* \subseteq P$ whose triples manifest as consistent displacement vectors in the embedding space.

Definition (Trajectory). For a triple $(s, p, o) \in \mathcal{K}$, the trajectory is the displacement vector $\mathbf{g}_{s,p,o} = f(o) - f(s)$, connecting the subject's embedding to the object's embedding.

Definition (Operation Consistency). For a predicate p with triples $\{(s_1, p, o_1), \dots, (s_n, p, o_n)\}$, the operation vector is $\mathbf{d}_p = \frac{1}{n} \sum_{i=1}^n \mathbf{g}_{s_i, p, o_i}$. The consistency of p is the mean cosine alignment of individual trajectories with the operation vector:

$$\text{consistency}(p) = \frac{1}{n} \sum_{i=1}^n \cos(\mathbf{g}_{s_i, p, o_i}, \mathbf{d}_p)$$

A predicate with consistency > 0.5 is a discovered operation: its triples are approximated by a single vector displacement.

3.2 Data Pipeline

1. Entity Import. Breadth-first search from a seed entity (Englishiki, Q1342448) through Wikidata, importing 500 entities with all their triples and linked entities (14,796 items total).

2. Embedding. Each entity's English label is embedded using mx-bai-embed-large (1024-dim) via Ollama. Aliases receive separate embeddings. Total: 41,725 embeddings.

3. Trajectory Computation. For each entity-entity triple, compute the displacement vector between subject and object label embeddings. Total: 16,893 entity-entity triples across 1,472 unique predicates.

3.3 Discovery Procedure

For each predicate p with ≥ 10 entity-entity triples:

1. Compute all trajectories $\{\mathbf{g}_i\}$
2. Compute operation vector $\mathbf{d}_p$ (mean trajectory)
3. Compute consistency: mean alignment of each $\mathbf{g}_i$ with $\mathbf{d}_p$
4. Compute pairwise consistency: mean cosine similarity between all pairs of trajectories
5. Compute magnitude coefficient of variation: stability of trajectory lengths

3.4 Prediction Evaluation

For each discovered operation ($\text{consistency} > 0.5$), we evaluate prediction accuracy using leave-one-out:

For each triple (s, p, o) :

1. Compute $\mathbf{d}_p^{(-i)} = \text{mean displacement excluding this triple}$
2. Predict: $\hat{o} = f(s) + \mathbf{d}_p^{(-i)}$
3. Rank all entities by cosine similarity to $\hat{o}$
4. Record the rank of the true object o

We report Mean Reciprocal Rank (MRR) and Hits@k for $k \in \{1, 5, 10, 50\}$.

3.5 Composition Test

To test whether operations can be chained, we find all two-hop paths $s \xrightarrow{p_1} m \xrightarrow{p_2} o$ where both p_1 and p_2 are discovered operations. We predict:

$$\hat{o} = f(s) + \mathbf{d}_{p_1} + \mathbf{d}_{p_2}$$

and evaluate whether the true o appears in the top-k nearest neighbors. We test 5,000 compositions.

4. Results

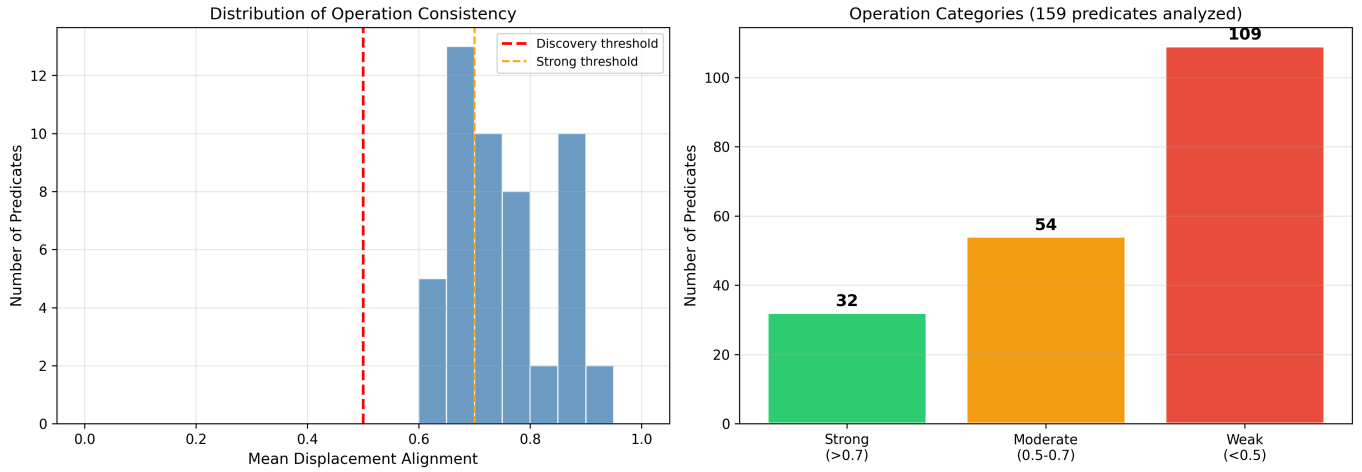


Figure 2: Distribution of predicate consistency scores. Left: histogram of alignment values. Right: category breakdown (32 strong, 54 moderate, 109 weak out of 159 analyzed).

4.1 Operation Discovery

Of 159 predicates with ≥ 10 triples, 86 (54.1%) produce consistent displacement vectors:

Category	Count	Alignment Range
Strong operations	32	> 0.7
Moderate operations	54	$0.5 - 0.7$
Weak/no operation	73	< 0.5

Table 1. Distribution of discovered operations by consistency.

The top 15 discovered operations:

Predicate	Label	N	Alignment	Pairwise	MagCV	Cos Dist
P8324	funder	25	0.930	0.859	0.079	0.447
P2633	geography of topic	18	0.910	0.819	0.097	0.200
P9241	demographics of topic	21	0.899	0.799	0.080	0.215
P2596	culture	16	0.896	0.790	0.063	0.202
P5125	Wikimedia outline	20	0.887	0.777	0.089	0.196
P7867	category for maps	29	0.878	0.763	0.099	0.205
P8744	economy of topic	30	0.870	0.749	0.094	0.182
P1740	cat. for films shot here	18	0.862	0.728	0.121	0.266
P1791	cat. for people buried here	13	0.857	0.714	0.121	0.302
P1465	cat. for people who died here	29	0.857	0.725	0.124	0.249
P163	flag	31	0.855	0.723	0.123	0.208
P2746	production statistics	11	0.850	0.696	0.048	0.411
P1923	participating team	32	0.831	0.681	0.042	0.387
P1464	cat. for people born here	32	0.814	0.653	0.145	0.265
P237	coat of arms	21	0.798	0.620	0.138	0.268

Table 2. Top 15 discovered operations by consistency (alignment with mean displacement). N = number of triples. Pairwise = mean cosine similarity between all pairs of trajectories. MagCV = coefficient of variation of trajectory magnitudes. Cos Dist = mean cosine distance between subject and object.

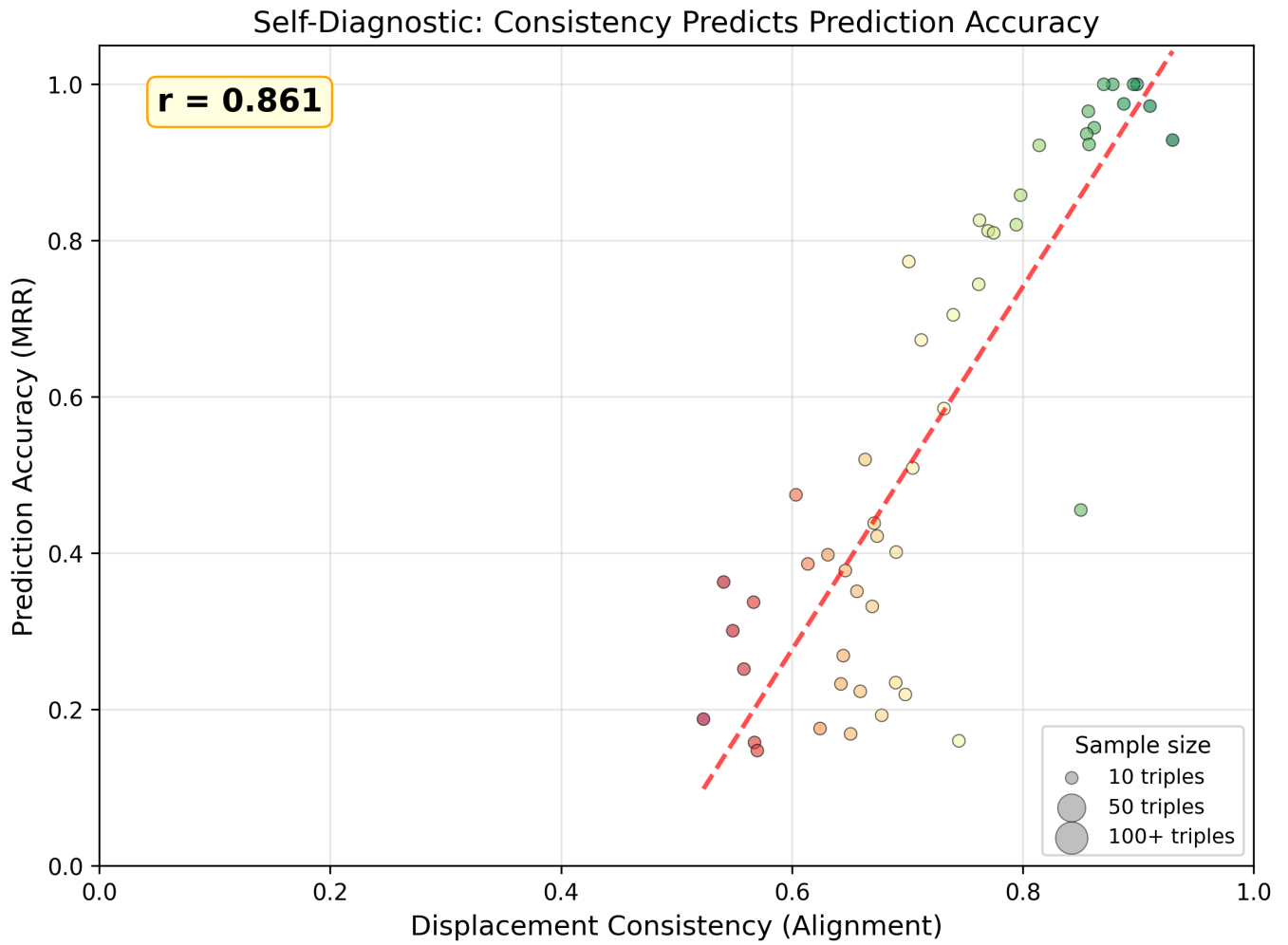


Figure 1: Displacement consistency (alignment) vs prediction accuracy (MRR). Each point is a discovered operation. Color indicates alignment strength. Point size indicates number of triples. $r = 0.861$, 95% CI [0.773, 0.926].

4.2 Prediction Accuracy

Leave-one-out evaluation of all 86 discovered operations:

Predicate	Label	N	Align	MRR	H@1	H@10	H@50
P9241	demographics of topic	21	0.899	1.000	1.000	1.000	1.000
P2596	culture	16	0.896	1.000	1.000	1.000	1.000
P7867	category for maps	29	0.878	1.000	1.000	1.000	1.000
P8744	economy of topic	30	0.870	1.000	1.000	1.000	1.000
P5125	Wikimedia outline	20	0.887	0.975	0.950	1.000	1.000
P2633	geography of topic	18	0.910	0.972	0.944	1.000	1.000
P1465	cat. for people who died here	29	0.857	0.966	0.966	0.966	0.966
P163	flag	31	0.855	0.937	0.903	0.968	1.000
P8324	funder	25	0.930	0.929	0.920	0.960	0.960
P1464	cat. for people born here	32	0.814	0.922	0.906	0.938	0.938
P237	coat of arms	21	0.798	0.858	0.762	0.952	1.000
P21	sex or gender	91	0.674	0.422	0.121	0.945	0.989
P27	country of citizenship	37	0.690	0.401	0.162	0.892	0.973

Table 3. Prediction results for selected operations (full table in supplementary). MRR = Mean Reciprocal Rank. H@k = Hits at rank k.

Aggregate statistics across all 86 operations:

Metric	Value
Mean MRR	0.350
Mean Hits@1	0.252
Mean Hits@10	0.550
Mean Hits@50	0.699

```
| Correlation (alignment ? MRR) | r = 0.780, p < 0.001 |
| Correlation (alignment ? H@1) | r = 0.791, p < 0.001 |
```

Table 4. Aggregate prediction statistics.

The correlation between displacement consistency and prediction accuracy ($r = 0.78$) is the central methodological finding: the discovery metric is also the quality metric. A predicate's geometric consistency, computable without any held-out evaluation, predicts how well that predicate will function as a vector operation.

4.3 Two-Hop Composition

Over 5,000 tested two-hop compositions ($S + d? + d?$):

```
| Metric | Value |
| Hits@1 | 0.058 (288/5000) |
| Hits@10 | 0.283 (1414/5000) |
| Hits@50 | 0.479 (2396/5000) |
| Mean Rank | 1029.8 |
```

Table 5. Two-hop composition results.

Selected successful compositions (Rank ≤ 5):

```
| Chain | Rank |
| Tadahira ->[citizenship]-> Japan ->[history of topic]-> history of Japan | 1 |
| Tadahira ->[citizenship]-> Japan ->[flag]-> flag of Japan | 1 |
| Tadahira ->[citizenship]-> Japan ->[cat. people buried here]-> Category:Burials in Japan | 2 |
| Tadahira ->[citizenship]-> Japan ->[cat. people who died here]-> Category:Deaths in Japan | 2 |
| Tadahira ->[citizenship]-> Japan ->[cat. associated people]-> Category:Japanese people | 3 |
| Tadahira ->[citizenship]-> Japan ->[head of state]-> Emperor of Japan | 4 |
| Tadahira ->[sex or gender]-> male ->[main category]-> Category:Male | 5 |
```

Table 6. Successful two-hop compositions. Note: all examples involve Fujiwara no Tadahira because our dataset is seeded from Englishiki (Q1342448), a Japanese historical text. Tadahira is one of the most densely connected entities in this neighborhood, appearing in many two-hop paths. The composition mechanism itself is general -- the examples reflect dataset composition, not a limitation of the method.

4.4 Failure Analysis

Predicates that resist vector encoding:

```
| Predicate | Label | N | Alignment | Pattern |
| P3373 | sibling | 661 | 0.026 | Symmetric |
| P155 | follows | 89 | 0.050 | Sequence (variable direction) |
| P156 | followed by | 86 | 0.053 | Sequence (variable direction) |
| P1889 | different from | 222 | 0.109 | Symmetric/diverse |
| P279 | subclass of | 168 | 0.118 | Hierarchical (variable depth) |
| P26 | spouse | 138 | 0.135 | Symmetric |
| P40 | child | 254 | 0.142 | Variable direction |
| P47 | shares border with | 197 | 0.162 | Symmetric |
| P530 | diplomatic relation | 930 | 0.165 | Symmetric |
| P31 | instance of | 835 | 0.244 | Too semantically diverse |
```

Table 7. Predicates with lowest consistency. Pattern = our characterization of why the displacement is inconsistent.

Three failure modes emerge:

1. Symmetric predicates (sibling, spouse, shares-border-with, diplomatic-relation): No consistent displacement direction because $f(A) - f(B)$ and $f(B) - f(A)$ are equally valid. Alignment ≈ 0 .
2. Sequence predicates (follows, followed-by): The displacement from "Monday" to "Tuesday" has nothing in common with the displacement from "Chapter 1" to "Chapter 2." The relationship type is consistent but the direction in embedding space is domain-dependent.
3. Semantically overloaded predicates (instance-of, subclass-of, part-of): "Tokyo is an instance of city" and "7 is an instance of prime number" produce wildly different displacement vectors because the predicate covers too many semantic domains.

Instance-of (P31) at 0.244 is particularly notable. It is the most important predicate in Wikidata (835 triples in our dataset) and a cornerstone of first-order logic, yet it does not function as a vector operation. This suggests that embedding spaces

systematically under-represent relational structure: the space encodes entities well but predicates poorly.

5. Discussion

5.1 What Makes an Operation Discoverable?

The pattern across Tables 2 and 7 is clear: discovered operations are functional relationships. Each country has one flag, one coat of arms, one head of state. These many-to-one mappings produce consistent displacements because the "direction from country to its flag" is a well-defined geometric direction. Symmetric or many-to-many relationships produce no consistent direction because there is no single "direction from person to their sibling."

This finding has implications for knowledge graph completion: vector arithmetic will succeed for functional predicates and fail for relational ones, regardless of the embedding model. The limitation is not in the model but in the nature of the relationship.

5.2 The Correlation as a Self-Diagnostic

The $r = 0.78$ correlation between consistency and prediction accuracy means the method is self-calibrating: you can determine which operations will work without needing ground-truth evaluation data. This is practically important because it means the method can be applied to embedding spaces where no labeled evaluation set exists -- the discovery process produces its own quality estimate.

Three Regimes of Embedding Space (with empirical evidence from mxbai-embed-large)

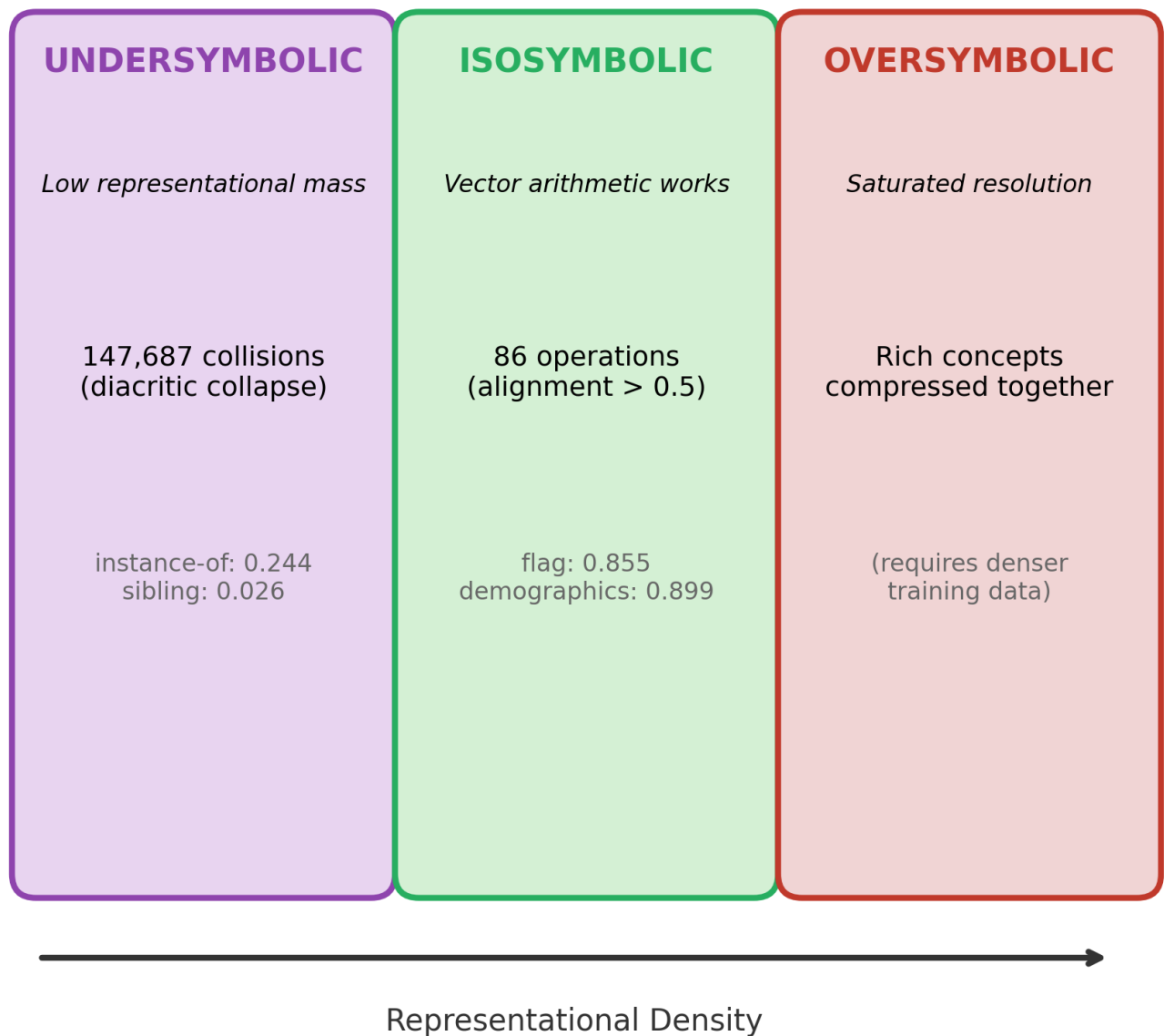


Figure 3: The three regimes of embedding space with empirical evidence. Undersymbolic: 147,687 collisions from diacritic collapse. Isosymbolic: 86 discovered operations. Oversymbolic: saturated resolution.

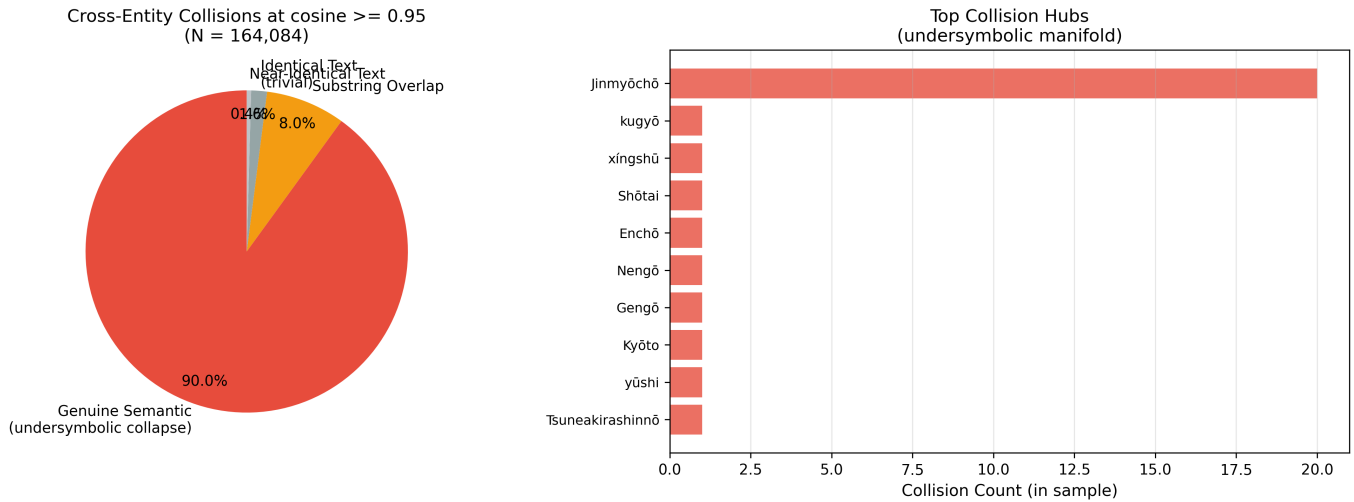


Figure 4: Collision type breakdown. 90% of 164,084 cross-entity collisions are genuine semantic (different text, near-identical embedding).

5.3 Three Regimes of Embedding Space

Our results, combined with collision analysis, reveal that embedding spaces are not uniformly structured. We identify three regimes:

- * **Oversymbolic regions** -- areas where the model compresses too many semantically rich concepts into overlapping coordinates. In an oversymbolic region, distinct and meaningful entities share embedding space because the model's representational capacity is saturated. This regime produces collisions between concepts the model has learned but cannot separate at the required granularity.

- * **Isosymbolic regions** -- the narrow manifold where vector arithmetic reliably preserves logical structure. Our 86 discovered operations live here. The functional predicates (flag, coat of arms, demographics) produce consistent displacements precisely because the entities involved are well-represented and well-separated in the embedding space. The isosymbolic regime is where our method works.

- * **Undersymbolic regions** -- sparse areas with insufficient representational mass to anchor specific concepts. These regions lack the training signal needed to differentiate their contents -- distinct inputs receive near-identical embeddings not because the model chose to group them, but because it never learned to distinguish them.

Empirical evidence: the Jinmyocho collapse. Our collision analysis finds 164,084 cross-entity embedding pairs with cosine similarity ≥ 0.95 . Of these, 147,687 (90%) are genuine semantic collisions: different text mapped to near-identical vectors. The collisions are dominated by romanized non-Latin-script terms -- the single text "Hokkaido" collides with 1,428 other entities, while "Jinmyocho" collides with 504 unique texts spanning romanized Japanese (kugyo, Shotai), Arabic (Djazaïr, Filastin), Irish (Eire), Brazilian indigenous languages (Aikanã, Amanaye), and IPA characters.

Crucially, this is an undersymbolic phenomenon, not an oversymbolic one. Tokenizer analysis reveals the mechanism: mxbai-embed-large's WordPiece tokenizer strips diacritical marks during normalization -- "Hokkaido" tokenizes to ['hokkaido'], "Tokyo" to ['tokyo'], "România" to ['romania']. Terms whose semantic content is carried primarily by diacritics lose that content at tokenization, collapsing into shared or similar subword sequences. The embedding space downstream of the tokenizer has no information to work with -- it maps these inputs to a degenerate neighborhood because it was never given the distinguishing features.

This resembles the glitch token phenomenon documented in LLM research (Li et al., 2024), where low-frequency tokens receive poorly trained embeddings. Our finding extends this to sentence-embedding models: it is not individual tokens but entire classes of input (romanized non-Latin scripts, diacritical text, IPA notation) that collapse into an undersymbolic manifold. The collision count -- 147,687 pairs -- quantifies the scale of this failure mode in a production embedding model.

This three-regime structure has practical implications. It suggests that vector arithmetic methods -- whether our trajectory displacement analysis or TransE-style learned translations -- will always be bounded by the topological quality of the underlying embedding space. No amount of algorithmic sophistication can extract consistent displacements from a region where distinct concepts have collapsed into the same coordinates. The undersymbolic collapse we document here is particularly insidious because it is invisible to standard evaluation: the model appears to embed these inputs normally, but the resulting vectors carry no discriminative information.

5.4 Implications for Neurosymbolic AI

Our results suggest a recategorization of the neurosymbolic landscape:

- * Constructive neurosymbolic: Build spaces that support logic (TransE, LTN, box embeddings)
- * Analytical neurosymbolic: Discover what logic exists in spaces not built for it (this work)

The analytical approach is complementary. It tells you what a given embedding space can do logically, without modification. This is useful for:

- * Model evaluation: Comparing embedding models by the richness of their latent logical structure
- * Hybrid architectures: Using vector arithmetic for operations that work (functional predicates) and falling back to symbolic graph traversal for operations that don't (relational predicates)
- * Embedding space cartography: Mapping which regions of a space support reliable vector arithmetic, identifying oversymbolic collapse zones, and diagnosing where a model needs more training data

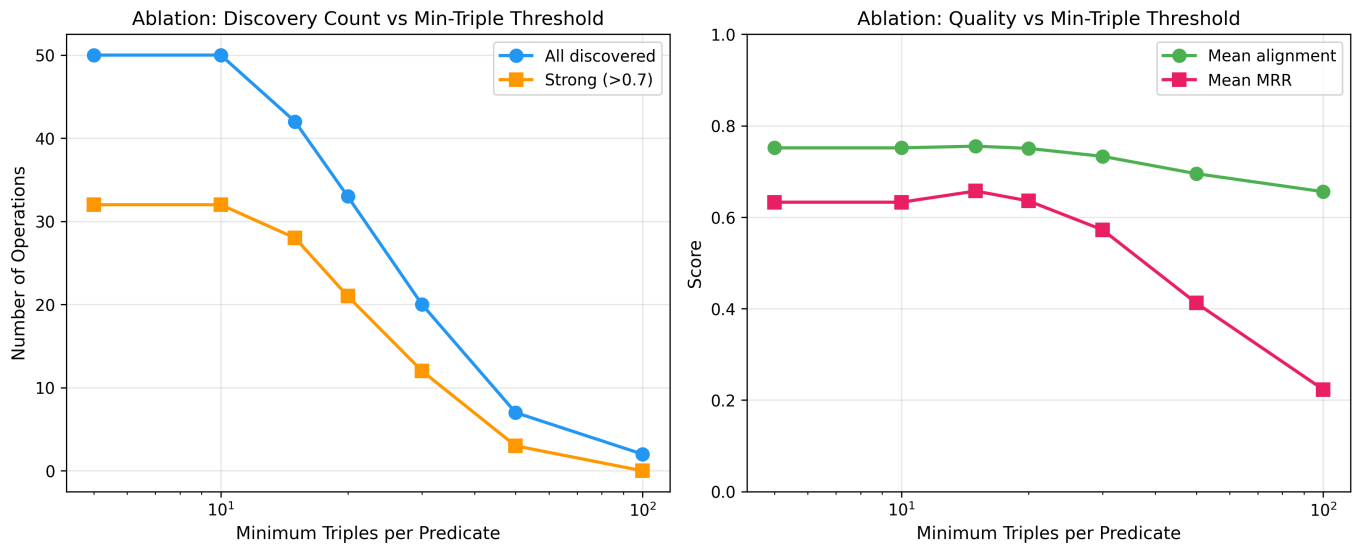


Figure 5: Ablation study. Discovery count and quality metrics vs minimum triple threshold. Results stable across thresholds 5-20.

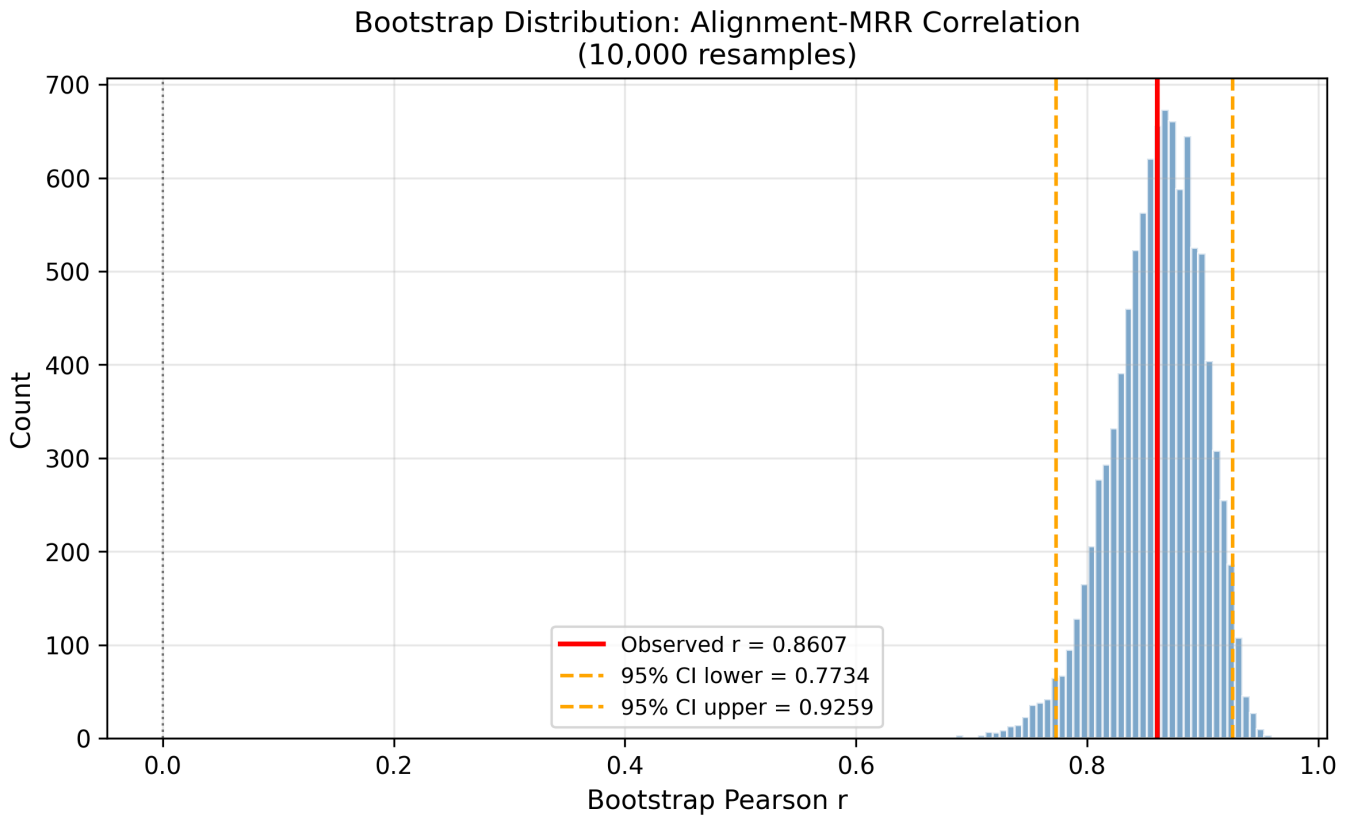


Figure 6: Bootstrap distribution of alignment-MRR correlation (10,000 resamples). 95% CI [0.773, 0.926], entirely above zero.

5.5 Limitations

1. Single embedding model. We test mxbai-embed-large only. The generality claim (arbitrary embedding spaces) requires testing on multiple models with different architectures and dimensionalities. The infrastructure supports this -- only the embedding function needs to change. Future work will evaluate cross-model generalization by applying the same analysis to models of varying size and training methodology, and test whether discovered transformation matrices transfer across embedding spaces -- which would establish a stronger form of model-agnostic neuro-symbolic reasoning.
2. Wikidata bias. Our entity set is seeded from Englishiki (Japanese historical text), producing a dataset heavy on Japanese history, linguistics, and geography. Different seeds would produce different distributions of predicates.
3. Label embeddings only. We embed entity labels (short text strings), not descriptions or full articles. Richer textual representations might improve results for predicates like instance-of.
4. No learned refinement. We report raw discovery results with no optimization. A learned projection layer on top of the discovered operations would likely improve prediction accuracy but would move toward the constructive paradigm.

6. Conclusion

We have shown that a general-purpose text embedding model, trained for semantic similarity with no logical training signal, latently encodes 86 first-order logical operations as consistent vector displacements. These operations can predict unknown triples (mean Hits@10 = 0.550), compose across two hops (Hits@10 = 0.283), and self-diagnose their reliability ($r = 0.78$ between consistency and accuracy). The method requires no training, no parameters, and no assumptions about the embedding geometry.

The key insight is not that embedding spaces can support logic -- TransE demonstrated that a decade ago -- but that they already do, without being asked. The logical structure is a byproduct of learning rich distributional representations. Our method excavates it.

All code is available at <https://github.com/EmmaLeonhart/Claw4S-submissions>. The full analysis reproduces end-to-end in approximately 30 minutes on commodity hardware with a local Ollama instance.

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